

# Lightning Talk

## Framework Laptops

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# Framework BoF + Hallway Track



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## What's Working?

- ▶ **Suspend/Resume Cycle (S0ix/S2)**
  - ▶ `hw.acpi.power_button_state=suspend_to_idle`
  - ▶ `hw.acpi.lid_switch_state=suspend_to_idle`
  - ▶ **4-line patch to ig(4) required**
- ▶ Trackpad gestures
  - ▶ 2-finger scroll
  - ▶ 3-finger swipe
- ▶ Trackpad multi-finger click
  - ▶ 2-finger = Right (CMenu)
  - ▶ 3-finger = Middle (Paste)
- ▶ **Touchscreen**



## S2 Idle Patch

- ▶ **Not Framework Specific**
- ▶ **Intel** specific
  - ▶ Low Power Sub-System (LPSS)
  - ▶ **Skylake or newer**
  - ▶ Including Alder/Raptor Lake-P
- ▶ Hardware **Lies**
  - ▶ Silicon never asserts `RESTORE_REQUIRED`
  - ▶ Fix in `ig4iic_resume()`
- ▶ **One small change**
  - ▶ **All** I2C-HID devices come back



# Desktop Environment (DE)

Below the Window Manager

## ▶ Volume

- ▶ Mute (Fn+F1)
- ▶ Down (Fn+F2)
- ▶ Up (Fn+F3)
- ▶ Output (Fn+F12)

## ▶ Brightness

- ▶ Down (Fn+F7)
- ▶ Up (Fn+F8)

## ▶ Prt Scr (Fn+F9)

- ▶ Take screenshot



## External Display Support

### ▶ Keyboard

- ▶ Screen Mirroring Key (Fn+F9) [Toggle]

### ▶ Window Manager (fvwm3)

- ▶ Screen Mirroring menu
- ▶ Screen Extension menu
  - ▶ Extend Right
  - ▶ Extend Left
  - ▶ Extend Above
- ▶ Recovery after hard disconnect
  - ▶ Laptop Display Only



# Framework 12 + FreeBSD 16

## Window Manager (fvwm3)

- ▶ **Super (Framework) key**
  - ▶ Alt+Esc emulation
  - ▶ Open Start Menu
- ▶ **Battery level indicator**
  - ▶ AC detection
  - ▶ fed by acpiconf -i 0



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## Limitations

- ▶ **Wake on AC / charge event**
  - ▶ BIOS setting ignored
  - ▶ AC unplug immediately wakes from s2idle
  - ▶ AC charge event also causes wake
  - ▶ No way to filter event in acpiec
  - ▶ **Firmware issue**





# Framework 12 + FreeBSD 16

## What's Next?

- ▶ **More trackpad gestures**
  - ▶ Pinch to zoom
- ▶ **More touchpad gestures**
  - ▶ Pinch to zoom
  - ▶ 2-finger scroll
  - ▶ 3-finger swipe



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# How we got here

### ▶ **Jun 13 (Sat) :: Baseline**

- ▶ Touch pad/screen dead after every resume

### ▶ **Jun 13-17 (Sat-Wed) :: Diagnosis**

- ▶ Traced silence past iichid down to ig4 bus
- ▶ Controller registers reading 0x0
- ▶ Every transfer fails with IIC\_ETIMEOUT

### ▶ **Jun 18 (Thu) :: Root-Cause Found**

- ▶ LPSS DEVICE\_IDLE never cleared on resume
- ▶ RESTORE\_REQUIRED not asserted on silicon
- ▶ Patched ig4iic\_resume()
- ▶ <https://reviews.freebsd.org/D57673>

